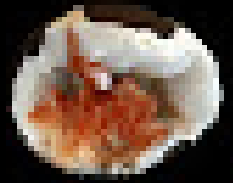
INDIAN MINERALS















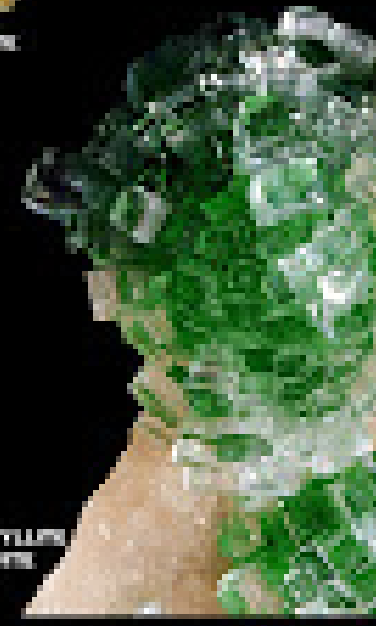
















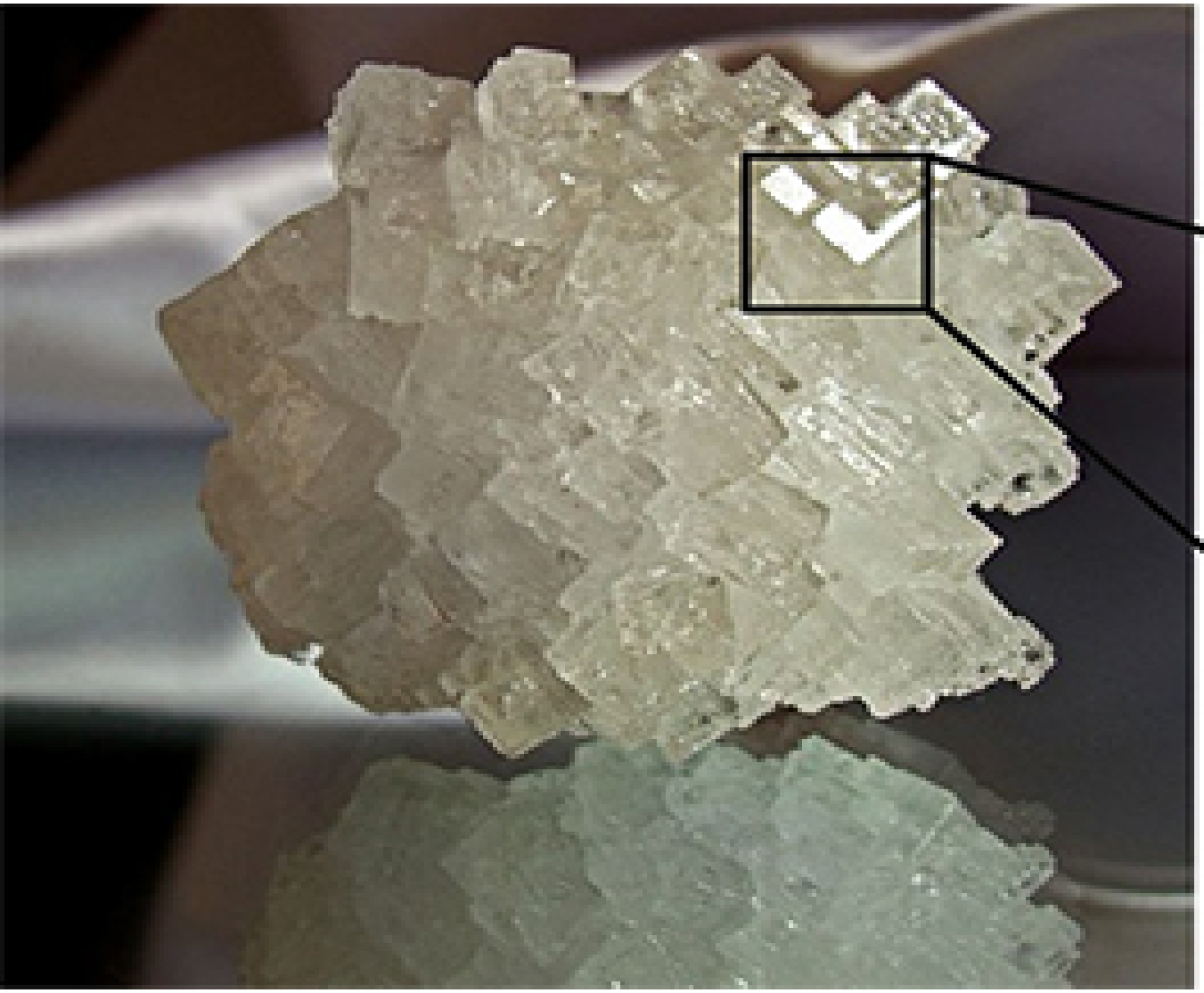


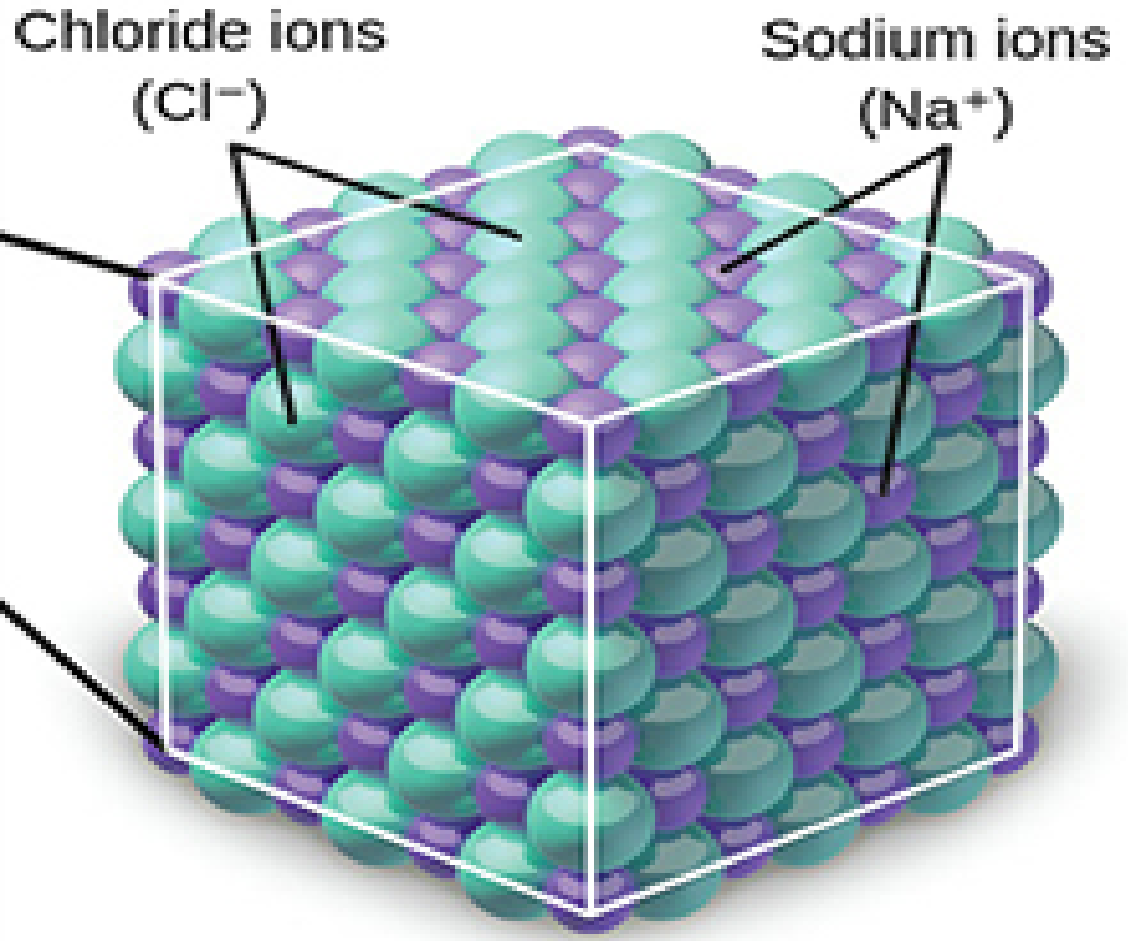


ix India Minerals

- Minerals are solid substances that are present in nature and can be made of one element or more elements combined together (chemical compounds).
- Gold, Silver and carbon are elements that form minerals on their own. They are called native elements. Instead, ordinary kitchen salt is a chemical compound that is called rock salt, which is a mineral formed of sodium and chlorine ions.

- Atoms, ions and molecules that form a mineral are present in the space in a tidy way and according to well-defined geometrical shapes, which are called crystal lattices. The structure of the crystal lattice defines the shape of the crystal as we see it.
- For example, rock salt or kitchen salt is a mineral formed of cubic-shaped crystals. Its crystal lattice has the same shape and consists of sodium and chlorine ions that are present in the space in alternate order.



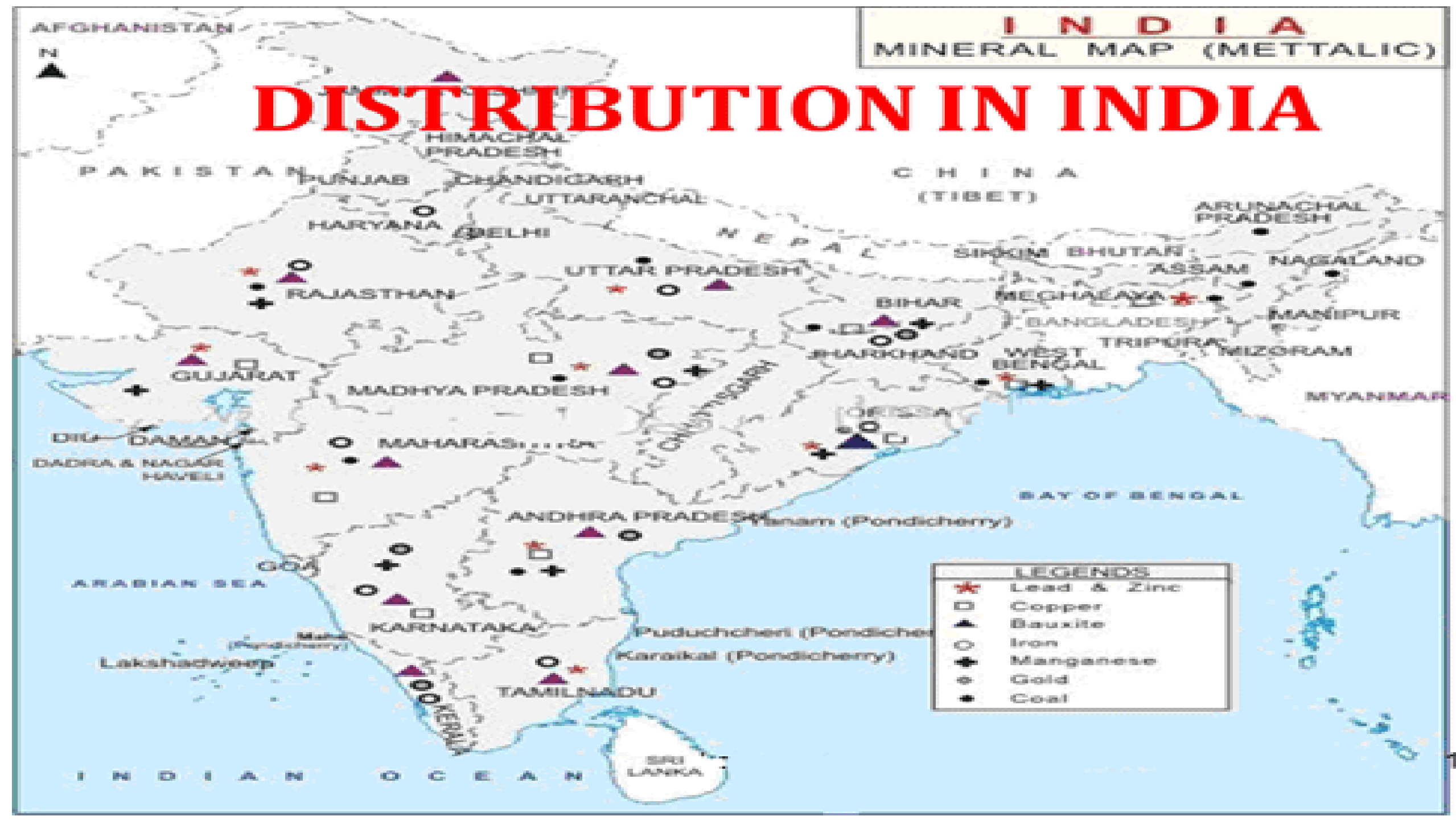


Sodium Chloride Crystal

- The order of atoms in the space and the way they combine with each other determine the way a mineral can laminate or exfoliate. Lamination is the property that some materials have to break according to their geometrical shapeAt the beginning of the scale there are very soft minerals that can be scratched with a nail, such as talc, chalk and calcite. At the end of the scale there is the diamond, which is the hardest mineral in nature

- Its chemical composition also determines the colour of the crystal, such as the yellow colour for the topaz, red for ruby, purple for amethyst quartz. Another characteristic of minerals is their hardness, which is their resistance to scratches. Hardness is classified by numbers (from 1 to 10), according to the Moh’s scale.

- India's major mineral resources include Coal (5th largest reserves in the world), Iron ore, Manganese ore (4 and 7th largest reserve and production in the world as in 2024), Mica (6th Largest), Bauxite (5th largest reserve in the world as in 2024), Chromite, Natural gas, Diamonds, Limestone and Thorium.


INDIA
MINERAL MAP (METTALIC)
DISTRIBUTION IN INDIA
AFGHANISTAN
N
PAKISTAN
HIMACHAL PRADESH
PUNJAB
CHANDIGARH
UTTARANCHAL
HARYANA
DELHI
CHINA
(TIBET)
NEPAL
ARUNACHAL PRADESH
SIKKIM
BHUTAN
ASSAM
NAGALAND
UTTAR PRADESH
RAJASTHAN
BIHAR
MEGHALAYA
MANIPUR
BANGLADESH
TRIPURA
MIZORAM
JHARKHAND
WEST BENGAL
GUJARAT
MADHYA PRADESH
CHHATTISGARH
ORISSA
MYANMAR
DIU
DAMAN
DADRA & NAGAR HAVELI
MAHARASHTRA
BAY OF BENGAL
ANDHRA PRADESH
Yanam (Pondicherry)
GOA
ARABIAN SEA
KARNATAKA
Puduchcheri (Pondicherry)
Mahe (Pondicherry)
Karaikal (Pondicherry)
Lakshadweep
TAMILNADU
KERALA
SRI LANKA
INDIAN OCEAN
LEGENDS
Lead & Zinc
Copper
Bauxite
Iron
Manganese
Gold
Coal

- India is one of the world's largest producers of mica, which is used by major global brands in the car and building sectors, electronics and electrical industries.
- Bihar, Jharkhand, Rajasthan and Andhra Pradesh are the top mica producing states in the country. "Giridih and Koderma districts of Jharkhand are rich in mica mineral.

- On the basis of chemical and physical properties, minerals are grouped as −
- Metallic minerals and
- Non-metallic minerals.
- Major examples of metallic minerals are iron ore, copper, gold, etc.
- Metallic minerals are further sub-divided as ferrous and non-ferrous metallic minerals.

- The minerals containing iron is known as ferrous and without iron is known as non-ferrous (copper, bauxite, etc.).
- Depending upon the origination, non-metallic minerals are either organic (such as fossil fuels also known as mineral fuels, which are derived from the buried animal and plant, e.g. such as coal and petroleum), or inorganic minerals, such as mica, limestone, graphite, etc.

- **Distribution of Minerals**
- Minerals are unevenly distributed on the earth's surface.
- All minerals are exhaustible in nature, i.e., will exhaust after a certain time.
- However, these minerals take long time to form, but they cannot be replenished immediately at the time of need.

- More than 97% of coal reserves occur in the valleys of Damodar, Sone, Mahanadi, and Godavari rivers.
- Petroleum reserves in India are located in the sedimentary basins of Assam, Gujarat, and Mumbai High (i.e. off-shore region in the Arabian Sea)

- **Mineral Belts in India**
- Further, there are three major mineral belts in India namely
- The North-Eastern Plateau Region,
- The South-Western Plateau Region, and
- The North-Western Region.

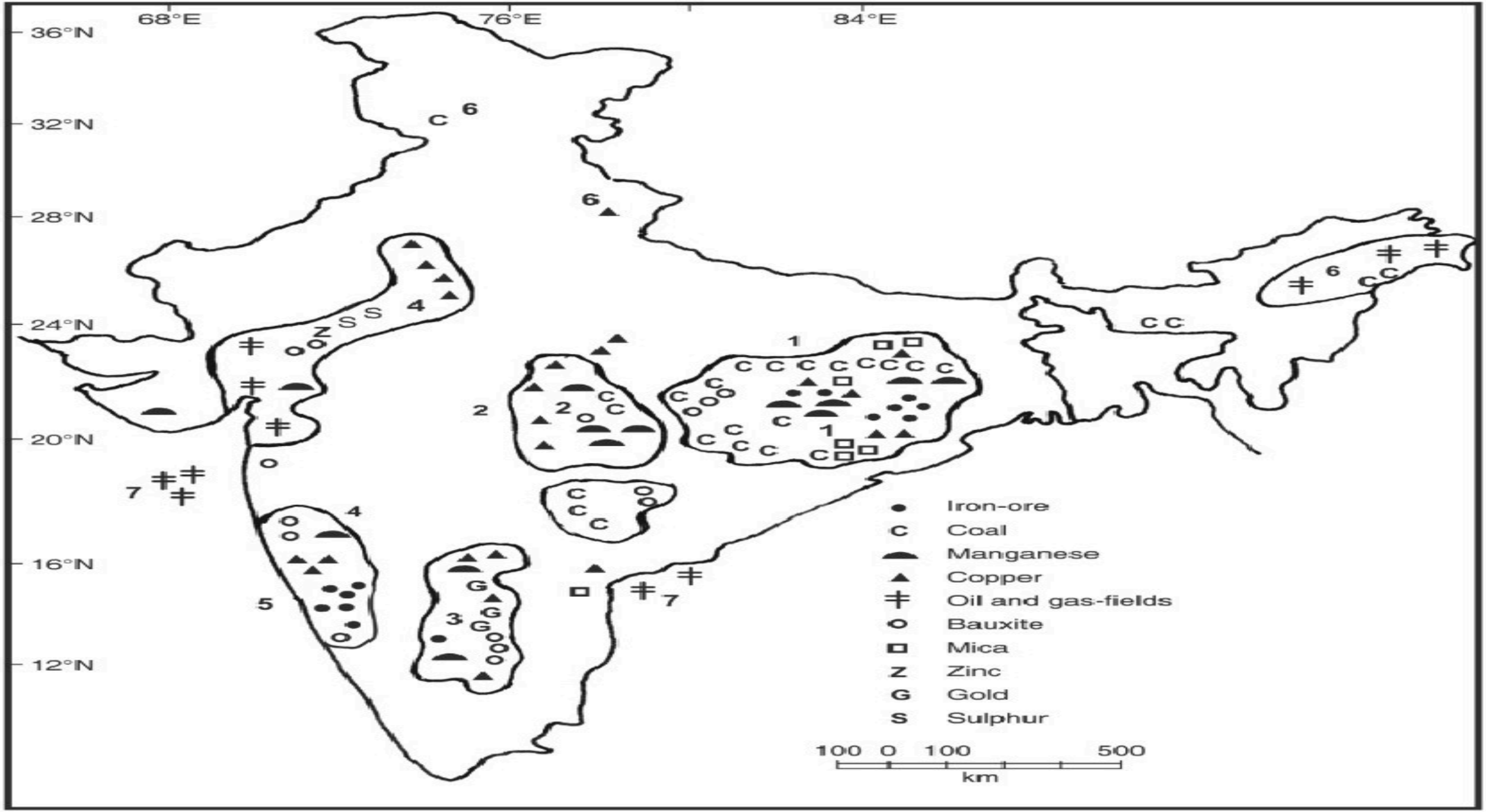


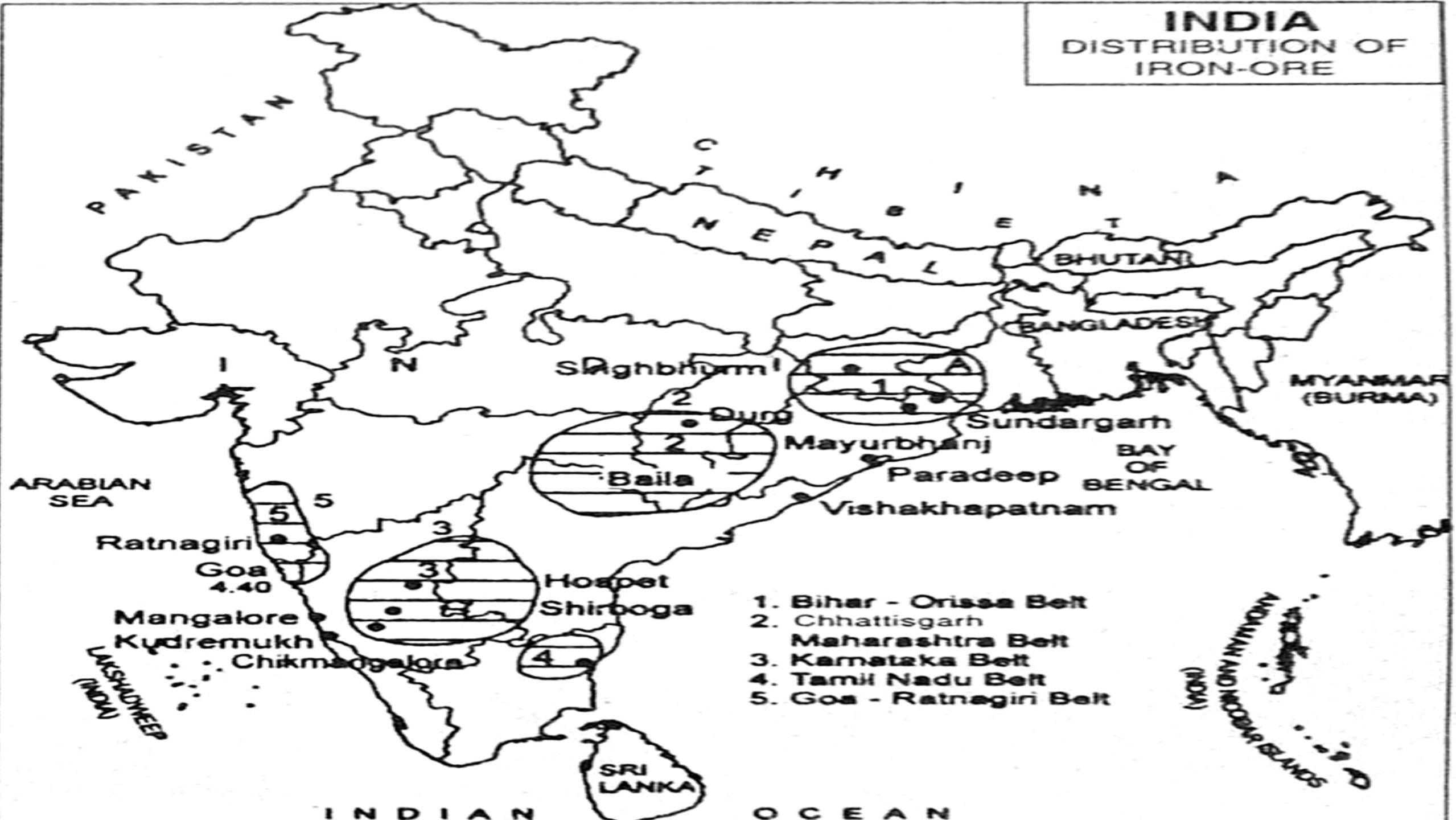
Fig. 7.1 Mineral Belts

- **North-Eastern Plateau Region**
- The major areas of north-eastern plateau region are Chhotanagpur (Jharkhand), Odisha, West Bengal, and parts of Chhattisgarh.
- Iron ore, coal, manganese, bauxite, and mica are the major minerals of the north-eastern plateau region.

- **South-Western Plateau Region**
- The south-western plateau region covers major parts of Karnataka, Goa, and contiguous Tamil Nadu uplands and Kerala.
- Major mineral resources of south-western plateau region are iron ore, manganese, and limestone.
- Kerala has deposits of monazite and thorium, and bauxite clay and Goa has deposits of iron ore.

- **North-Western Region**
- The north-western region covers the areas of Aravalli in Rajasthan and parts of Gujarat.
- Major minerals of north-western regions are copper and zinc; other significant minerals include sandstone, granite, and marble, along with Gypsum and Fuller’s earth deposits.

- In addition, Gujarat and Rajasthan, both have rich sources of salt.
- The Himalayan belt is also an important mineral belt, as it has rich deposits of copper, lead, zinc, cobalt, and tungsten.


INDIA
DISTRIBUTION OF
IRON-ORE
PAKISTAN
CHINA
NEPAL
BHUTAN
BANGLADESH
MYANMAR
(BURMA)
INDIA
Singhbhum
Durg
Baila
Mayurbhanj
Sundargarh
Paradeep
Vishakhapatnam
BAY
OF
BENGAL
ARABIAN
SEA
Ratnagiri
Goa
4.40
Hospet
Shimoga
Mangalore
Kudremukh
Chikmangalore
LAKSHADWEEP
(INDIA)
SRI
LANKA
INDIAN
OCEAN
ANDAMAN AND NICOBAR ISLANDS
(INDIA)
1. Bihar - Orissa Belt
2. Chhattisgarh
Maharashtra Belt
3. Karnataka Belt
4. Tamil Nadu Belt
5. Goa - Ratnagiri Belt

- **Iron**
- About 95% of total reserves of iron ore is found in the States of Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh, and Tamil Nadu.

- Sundergarh, Mayurbhanj, and Jhar are the major iron ore regions in Odisha and the important mines are Gurumahisani, Sulaipet, Badampahar (Mayurbhanj), Kiruburu (Kendujhar), and Bonai (Sundergarh).
- Noamundi (Poorbi Singhbhum) and Gua (Pashchimi Singhbhum) are important mines in Jharkhand.

- Dalli and Rajhara in Durg district are the important mines of Chhattisgarh.
- Sandur-Hospet area of Ballari district, Baba Budan hills, and Kudremukh in Chikkamagaluru district, and parts of Shivamogga are the important iron ore regions in Karnataka.

- The districts of Chandrapur, Bhandara, and Ratnagiri are the iron regions in Maharashtra.
- Other iron ore regions in India are Karimnagar and Warangal district of Telangana, Kurnool, Cuddapah, and Anantapur districts of Andhra Pradesh, and Salem and Nilgiris districts of Tamil Nadu

- **Manganese**
- Odisha is the leading producer of Manganese.
- Bonai, Kendujhar, Sundergarh, Gangpur, Koraput, Kalahandi, and Bolangir are the major manganese regions in Odisha.

- Dharwad, Bellary, Belagavi, North Canara, Shivamogga, Chitradurg, Tumkur, and Chikkmagaluru are major manganese regions in Karnataka.
- Nagpur, Bhandara, and Ratnagiri districts are the major regions of manganese in Maharashtra.
- Balaghat-Chhindwara-Nimar-Mandla, and Jhabua districts are the important manganese regions of Madhya Pradesh.

- **Coal**
- Coal deposits are primarily found in eastern and south-central India. Jharkhand, Odisha, Chhattisgarh, West Bengal, Madhya Pradesh, Telangana and Maharashtra accounted for 98.26% of the total known coal reserves in India.

- Coal resources of India are available in older Gondwana Formations of peninsular India and younger Tertiary formations of north-eastern region.

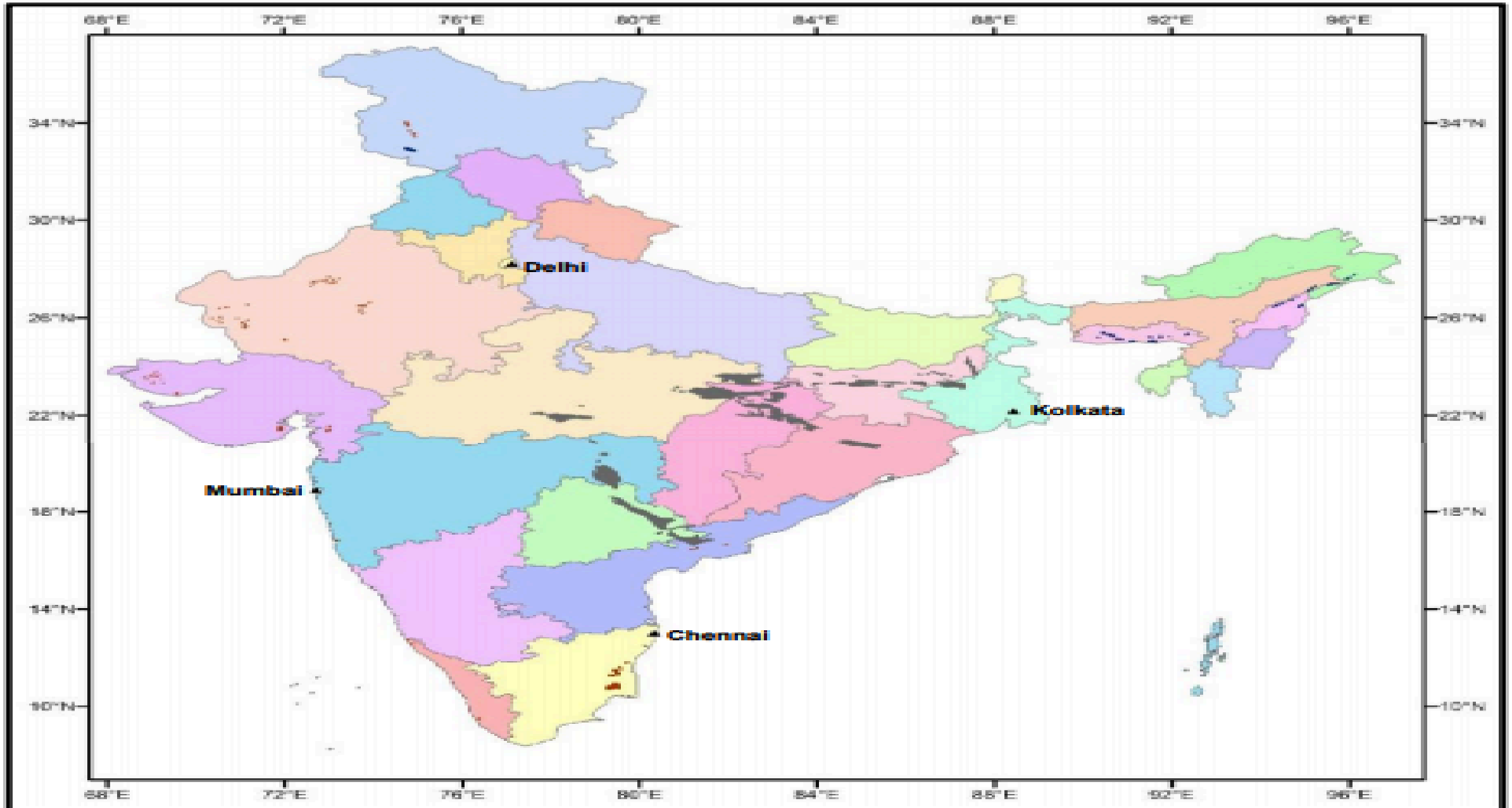
- Coal accounts for 67% of total commercial energy produced in India.
- Coal Gradations:
- 1. Anthracite (more than 80% carbon content)
- 2. Bituminous (60 to 80% carbon content)
- 3. Lignite (40 to 60% carbon content)
- 4. Peat (less than 40% carbon content)

- Major Coal Fields
- West Bengal: Raniganj (oldest coal field of India)
- Jharkhand: Jharia(The largest coal field in India), Bokaro, Giridih, Karanpura,Ramgarh, Daltonganj
- Madhya Pradesh:Singrauli, Suhagpur, Johilla, Umaria, Satpura coalfields

- Orissa : Talcher, Himgiri, Rampur
- Telangana: Godavari Khani, Singareni, Bellampalli
- Chhattisgarh: Korba, Bisrampur, Sonhat, Jhilmil, Hasdo-Arand

- Assam: Makum, Najira, Janji, Jaipur
- Meghalaya: Umralong, Darrangiri, Cheerapunji, Mawlong, Langrin

- Lignite is mainly found in Tamil Nadu, Rajasthan, Gujarat and a little in Jammu & Kashmir.



Distribution of coal and lignite fields of India

- **Petroleum or Mineral Oil**
- Major Fields: The mineral oil resources of India are distributed in three important basins.
- 1. Upper Assam or the Naharkatia-Moran Region: Major oil wells in this region are Digboi, Naharkatia, Moran, Lakwa, Sibsagar & Rudra-sagar. Assam valley is oldest oil field of India.

- 2. Mumbai High: This is an offshore source, lying 167 km to the north-west of Mumbai in Arabian Sea. This field is called Sagar Samrat. About 2/3rd of the total crude petroleum is produced from Mumbai High.
- 3. Cambay Basin: This basin lies in the State of Gujarat and major oil-wells are Ankleshwar, Kosamba, Kalol, Lunej, Dhalka, Mehsana, Nawagam and Sobhasan.

- **Natural Gas resource**
- Over 3/4th of the production of natural gas comes from the Mumbai High. Gujarat accounts roughly for 10% of gas production.

- Distribution of free gas resources: Natural gas in the free from is derived from the following regions:
- 1. The largest reserves come from the offshore Mumbai High
- 2. Cambay basin in Gujarat
- 3. Tripura
- 4. Kaveri offshore basin in Tamil Nadu
- 5. Andhra Pradesh
- 6. Thanot in Jaisalmer district of Rajasthan

- **Iron Ore**
- Iron ore are of four types:
- Magnetite (Black ore): Iron content is 72%
- Haematite (Red ore): Iron content is 70%
- Limonite (Brown ore): Iron content is 60%
- Siderite (Grey ore): Iron contest is 45%

- India is one of the best endowed countries in the magnitude and quality of iron ores. The country possesses over 20% of the world's total reserves, second largest after the reserves of the Commonwealth of Independent States (CIS).
- The quality of the Indian ores is very good.

- **Jharkhand-Orissa Belt:**
- In Jharkhand iron ore is found in the Barajamda group of mines covering apart from the Singhbhum districts.
- In Orissa, iron ore is found in the Gurumahisani and Badampahar group od mines in the Mayurbhanj district and in Keonjhar and Sundergarh districts.

- Chhattisgarh-Maharashtra (Durg-Bastar-Chandrapur Belt): In this region, iron ore is found in Dallirajhara in Durg district and Bailadila in Bastar district and in the Lohara – Pipalgaon – Surajgarh region of eastern Maharashtra.

- Karnataka: Here, iron ore is found in the Bellary-Chitradurga-Chikmanglur-Tumkur belt. Magnetite, which is extracted from the Dharwar and Cuddapah rocks, is found in Kudremukh.
- Tamil Nadu: Here, Iron ore is found in the Salem-Tiruchirapalli-North Arcot belt.
- Rajasthan: A rich tract has been discovered here in the Jaisalmer district.

- **Manganese**
- India has world’s second largest reserves of manganese ore next only to Zimbabwe. About 1/5th of the world’s deposits of manganese are found in India.

- India is fifth largest producer of manganese ore.
- Manganese ore is an important ingredient in the manufacture of iron and steel, and it is the basic raw material for manufacturing ferro-manganese alloy. These industries together account for over 84% of the total quantity kof manganese ore consumed in India.

- Orissa: The Gondite deposits occur in the Keonjhar and Sundergarh districts. The Kodurite and Khondolite deposits in the Koraput and Kalahandi districts. The Lateritic deposits in Bolangir and Sambalpur districts.
- Jharkhand: Manganese is found in Singhbhum district.

- Karnataka: Manganese ore is found in the districts of Bellary, Chitradurga Shimoga, Tumkur and North Kanara.
- M.P. & Chhatisgarh: Manganese is found in Balaghat and Chhindwara.
- Maharashtra: Bhandara and Nagpur in Maharashtra have manganese ore deposits.

- Andhra Pradesh: Srikakulam & Visakhapatnam.
- Gujarat: Vadodara and Panchmahal have the deposits.
- Rajasthan: Banswara

- **Alluminium (Bauxite)**
- Alluminium is mainly used in aluminium refractories, aircraft, automobile industry and the chemical industry. Although extraction of aluminium is a power-intensive process, aluminium is fast replacing steel in industries.

- The bauxite deposits are mainly spread over the Eastern Ghats.
- M.P. & Chhattisgarh: Nearly 30% of the country’s production of bauxite comes from Madhya Pradesh. Bauxite deposits are found in Amarkantak plateau in Shahdol district, Maikala hills and the plateau region of Sarguja-Bilaspur-Rajgarh-Katni in Chattisgarh.

- Jharkhand: Ranchi and Palamu
- Gujarat: Jamnagar, Kaira, Sabarkantha, Surat and Kachchh.
- Karnataka: Bauxite reserves are found in Belgaum
- Maharashtra: Kolaba, Kolhapur and Ratnagiri

- Tamil Nadu: Salem, Nilgiri, Coimbatore and Madurai
- Uttar Pradesh: Bauxite is extracted in Uttar Pradesh in Banda.
- Jammu and Kashmir: Here, high grade bauxite deposits are found in Poonch and Udhampur.

- **Gold**
- There are three important gold fields in the country:
- 1.Kolar gold fields in Kolar district (Karnataka)
- 2. Hutti gold fields in Raichur district (Karnataka)
- 3. Ramgiri gold fields in Anantpur district (Andhra Pradesh)

- **Lead and Zinc**
- Distribution:
- Rajasthan: Lead and Zinc are extracted from the Zawar mines in Udaipur district and from Anguncha-Rampura in Bhilwara district. In these mines, silver ore is also extracted.
- Andhra Pradesh: Kadapa
- Gujarat: Lead and Zinc are extracted from Banaskantha, Vadodara, Panchmahal and Surat.

- **Mica**
- Due to its excellent di-electric strength, low power loss factor, insulating properties and resistance to high voltage, mica is one of the indispensable minerals used in electric and electronic industries. India is the leading producer of sheet mica and almost the whole of it is exported mica.

- India accounts for 60% of the world trade. International demand however has fallen because of mica substitutes like plastics and synthetics gaining importance. This has led to a decline in production of mica over recent decades.

- Mica is produced mainly in the states of Jharkhand, Bihar, Rajasthan and Andhra Pradesh.
- Jharkhand & Bihar: Here, mica is extracted from the Gaya-Hazaribagh belt, with Koderma forest area contributing a large part of the country's total production.
- Rajasthan: Here, mica is mined in the Beawar, Ajmer belt, Banswara-Dungarpur belt, Bhilwara, Tonk and Kaunthal in Udaipur.
- Andhra Pradesh: Here, mica is extracted from Nellore in Gudur district.

- **Copper**
- Jharkhand & Bihar: The Signhbhum district of Jharkhand has the largest copper ore reserves in Asia. Other districts having copper ore reserves are Hazaribagh, Santhal Parganas, Gaya and Palamu.

- Rajasthan: Khetri belt (oldest copper mine of India) which includes the Singhana area in Jhunjhunu district and Kho-Dariba area in the Alwar district. Besides these, copper is also mined in Delwara-Kerovli in Udaipur and Aguncha-Rampur in Bhilwara.

- M.P. & Chhattisgarh: Balaghat and recently, reserves have been discovered in Malanjkhand also.
- Andhra Pradesh:Guntur and Kurnool
- Karnataka: Chitradurga and Hassan
- Maharashtra: Chandrapur belt.